CYBERSTER BLUE TEAM

INTERNSHIP PROGRAM | BATCH #02

WEEK 1 REPORT

PHASE ONE - SOC Operations & SIEM Foundations

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1. INTRODUCTION TO BLUE TEAM & SOC

1.1 What is Blue Team?

Team	Role	Focus
🔵 Blue Team	Defenders	Protect systems, monitor threats, respond to incidents
🔴 Red Team	Attackers	Simulate attacks, find vulnerabilities, test defenses
🟣 Purple Team	Collaboration	Bridge between Red & Blue, improve overall security

1.2 What is SOC (Security Operations Center)?

A Security Operations Center (SOC) is a centralized unit that monitors, detects, analyzes, and responds to cybersecurity incidents in real-time.

5 Key SOC Functions:

✅ Monitoring ✅ Detection ✅ Analysis ✅ Response ✅ Reporting

1.3 About Cyberster Internship

Program Details	Information
Program Name	Cyberster Blue Team Internship
Batch	#02
Phase One	SOC Foundations & SIEM Setup
Phase Two	Advanced Threat Hunting
Duration	8 Weeks
Primary Tool	Wazuh SIEM

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2. NETWORKING FUNDAMENTALS

2.1 OSI Model (7 Layers)

Layer #	Layer Name	Function	Examples
7	Application	User interaction with network services	HTTP, FTP, DNS, SMTP
6	Presentation	Data formatting, encryption, compression	SSL/TLS, JPEG, ASCII
5	Session	Connection management and control	NetBIOS, RPC, PPTP
4	Transport	End-to-end communication, reliability	TCP, UDP
3	Network	Routing and logical addressing	IP, ICMP, OSPF
2	Data Link	Physical addressing (MAC), error detection	Ethernet, MAC, Switch
1	Physical	Raw bit transmission over physical medium	Cables, Wi-Fi, Hubs

2.2 TCP/IP Model (4 Layers)

Layer	Function	Protocols
Application	User services and applications	HTTP, HTTPS, FTP, DNS, SMTP, SSH
Transport	Host-to-host communication	TCP, UDP
Internet	Routing and addressing	IP, ICMP, ARP
Network Access	Physical transmission	Ethernet, Wi-Fi

2.3 Important Ports for SOC Analysts

Port	Protocol	Service	Purpose
22	SSH 🔐	Secure Shell	Secure remote access
25	SMTP 📧	Simple Mail Transfer Protocol	Email sending
443	HTTPS 🔒	HTTP Secure	Encrypted web traffic
80	HTTP 🌐	Hypertext Transfer Protocol	Unencrypted web traffic
445	SMB 📁	Server Message Block	Windows file sharing
3389	RDP 💻	Remote Desktop Protocol	Windows remote access
1514	Wazuh	Agent Communication	UDP
1515	Wazuh	Agent Communication	TCP

Key Takeaway: Monitoring these ports is critical for detecting unauthorized access and lateral movement.

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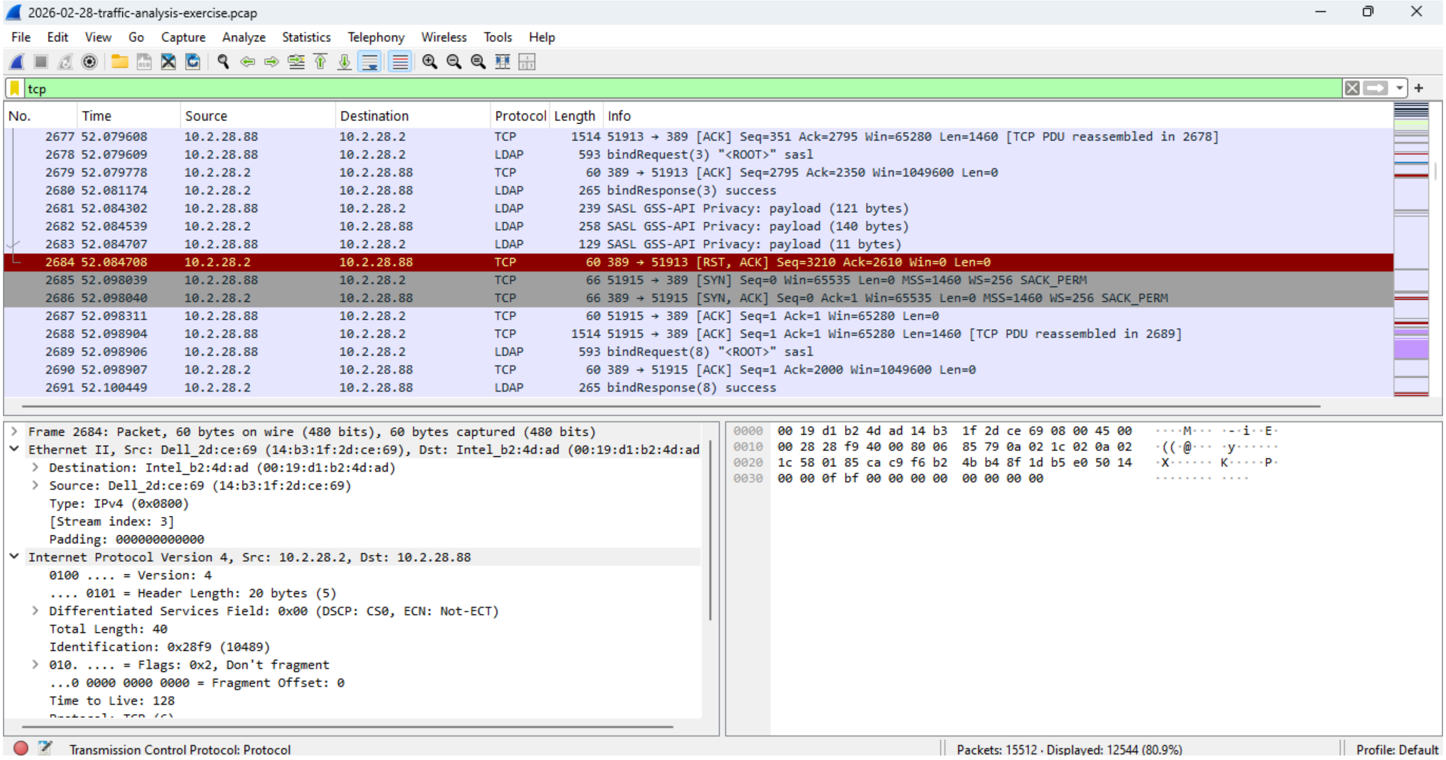
3. WIRESHARK TRAFFIC ANALYSIS

3.1 What is Wireshark?

Wireshark is an open-source network protocol analyzer used for capturing and analyzing network traffic.

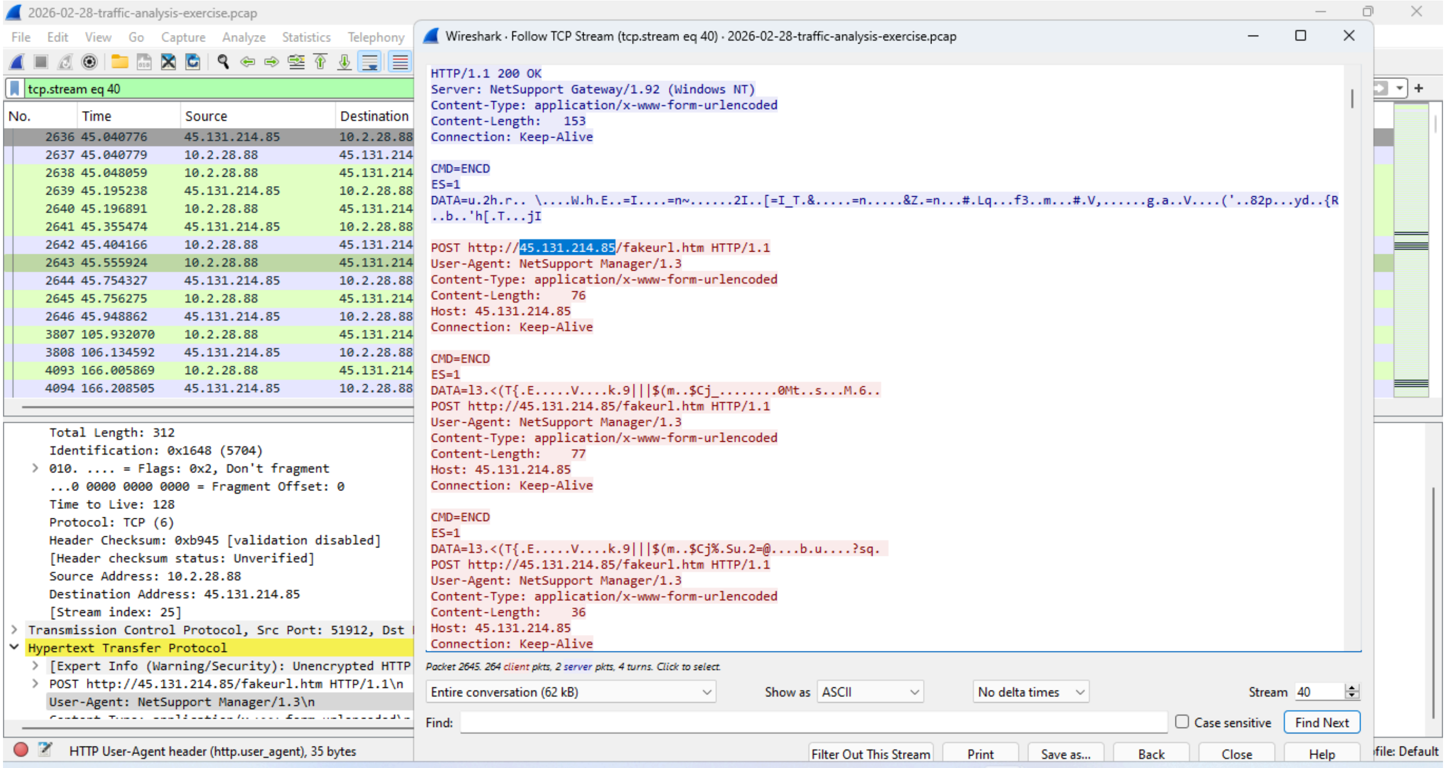
SOC Analysts Use Wireshark For:

• Packet Capture • Protocol Analysis • Threat Detection • Incident Investigation

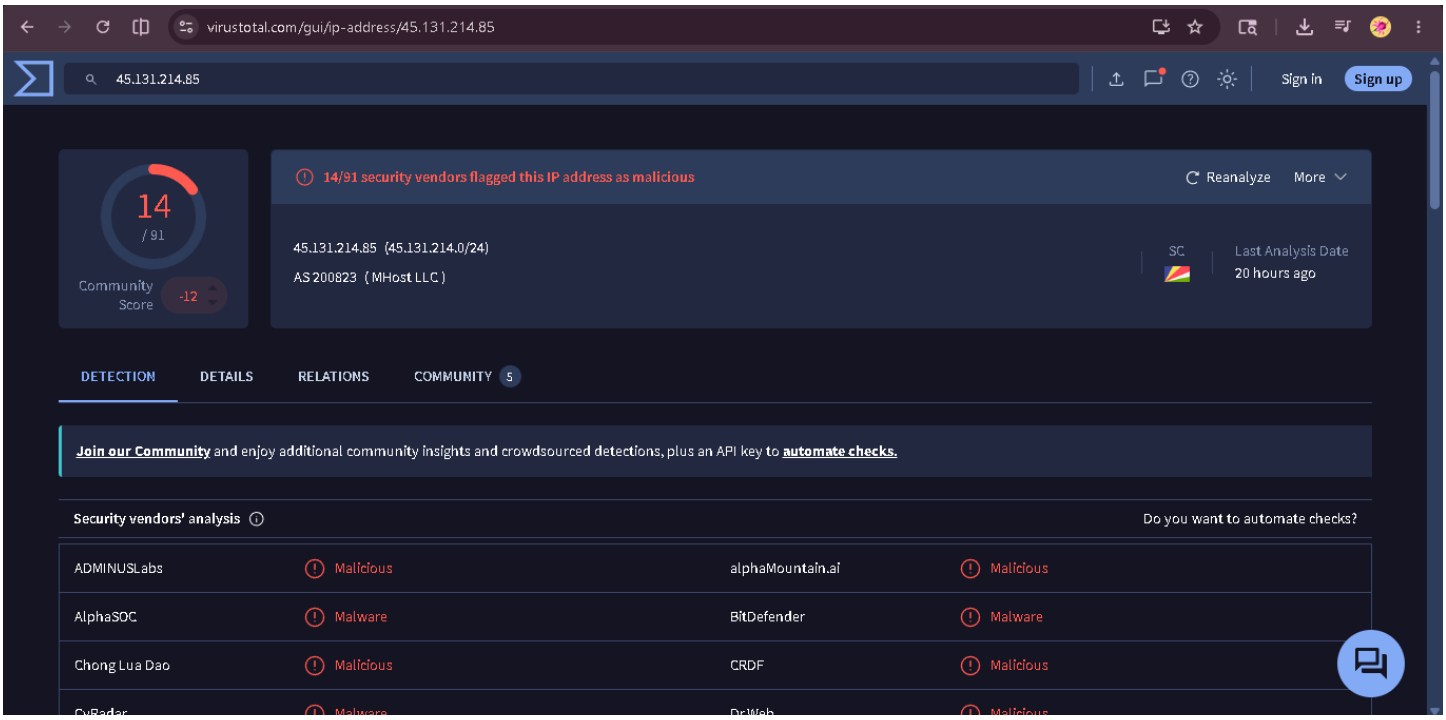


3.2 Suspicious TCP Traffic Detection

Attribute	Value
Source IP	45.131.214.85
Destination IP	10.2.28.88
ASN	AS200823 (MHost LLC)
Virus Total Detection	🔴 14/91 vendors flagged
Community Score	-12 (Malicious)
Threat Level	HIGH



Suspicious TCP Traffic Stream Follow



VirusTotal Detection Results

3.3 TCP Endpoints Analysis

IP Address	Packets	Bytes	Role
10.2.28.88	12,544	6 MB	🔴 Victim
10.2.28.2	4,098	1 MB	🟡 Gateway
45.131.214.85	—	—	🔴 Malicious External

Conclusion: IP 45.131.214.85 flagged as malicious. SOC would block and isolate affected endpoint.

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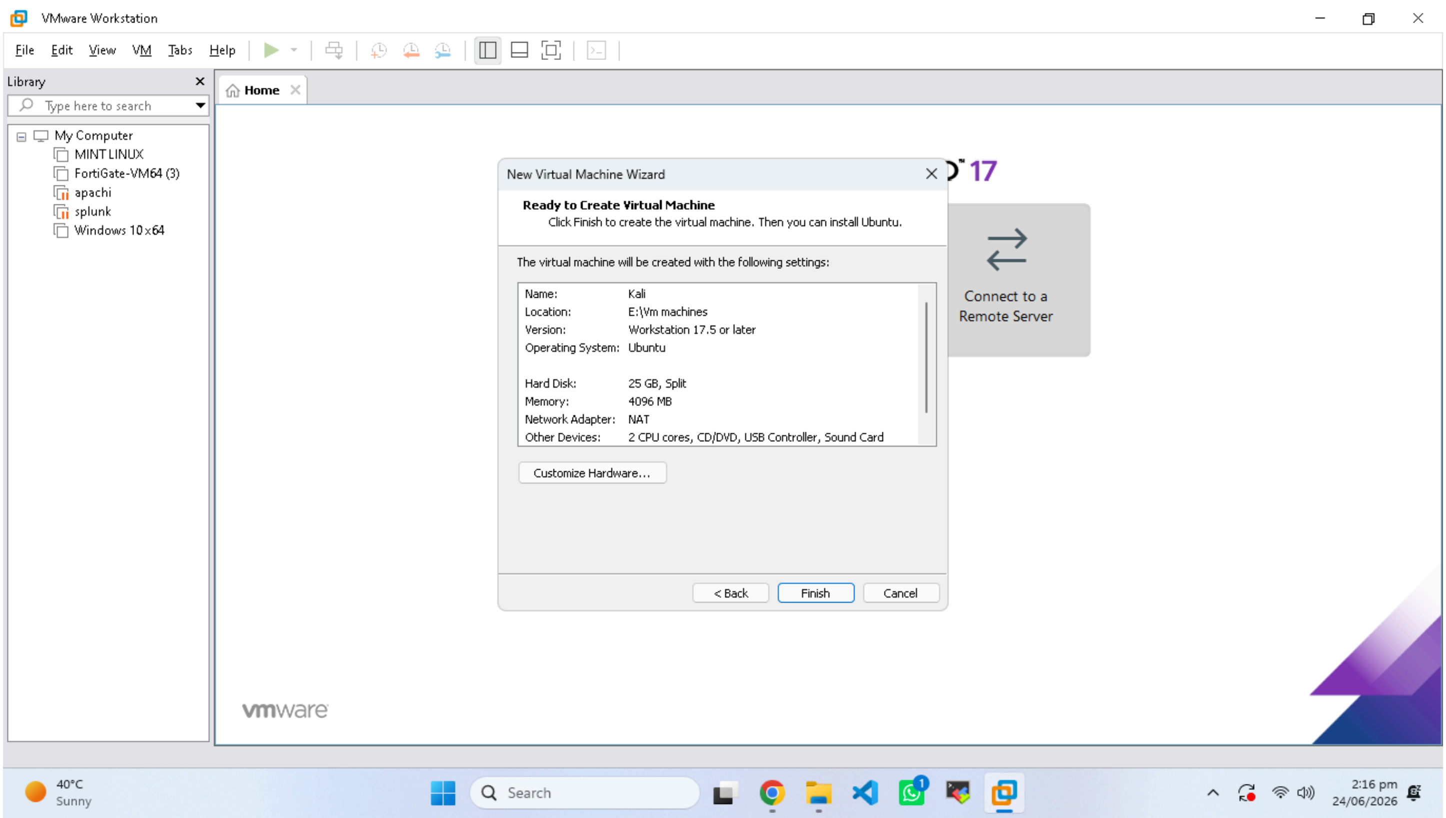
4. VIRTUAL LAB SETUP (VMware)

4.1 Why a Virtual Lab?

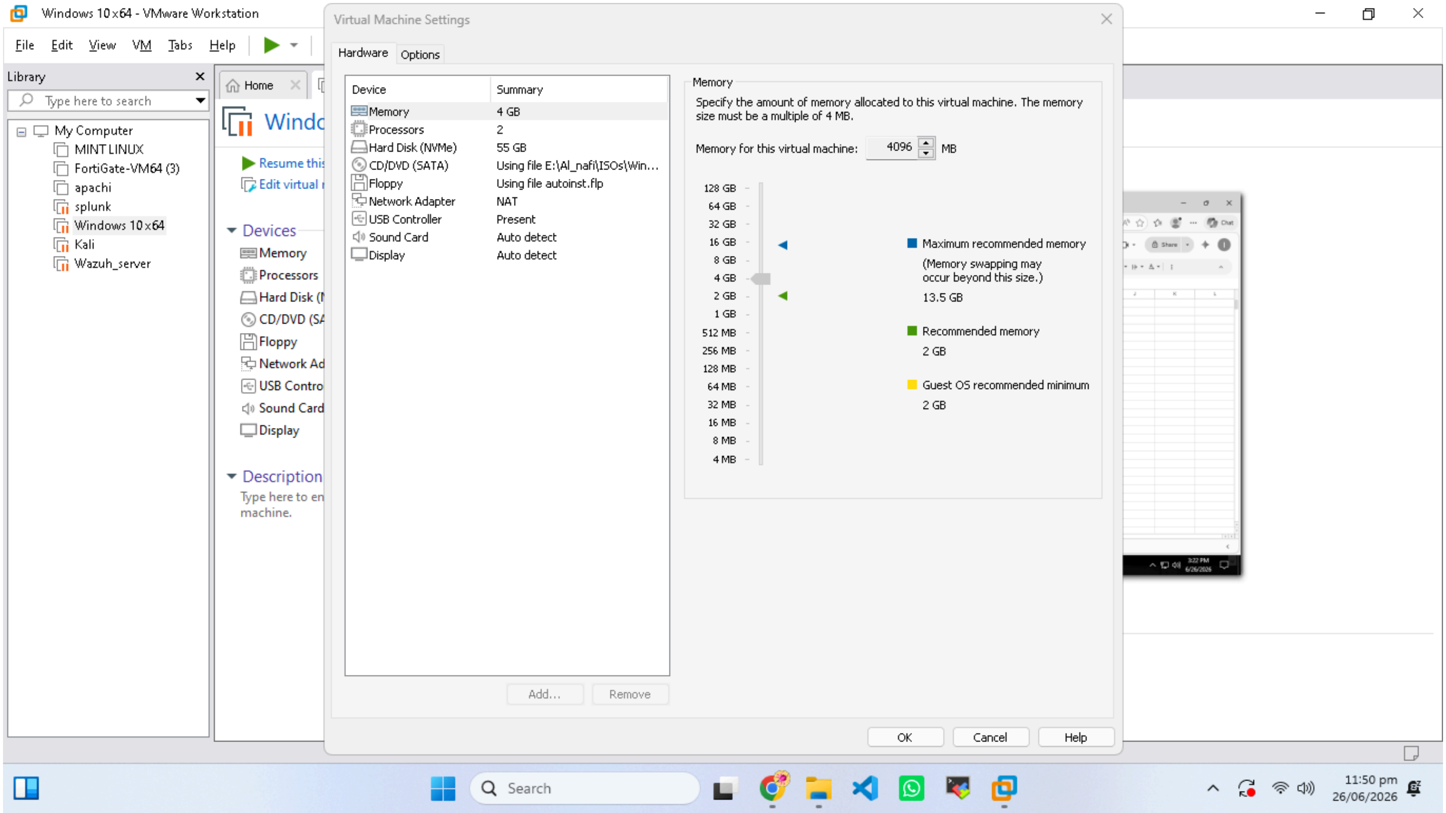
✅ Isolation ✅ Flexibility ✅ Cost-Effective ✅ Realistic

4.2 Lab Architecture

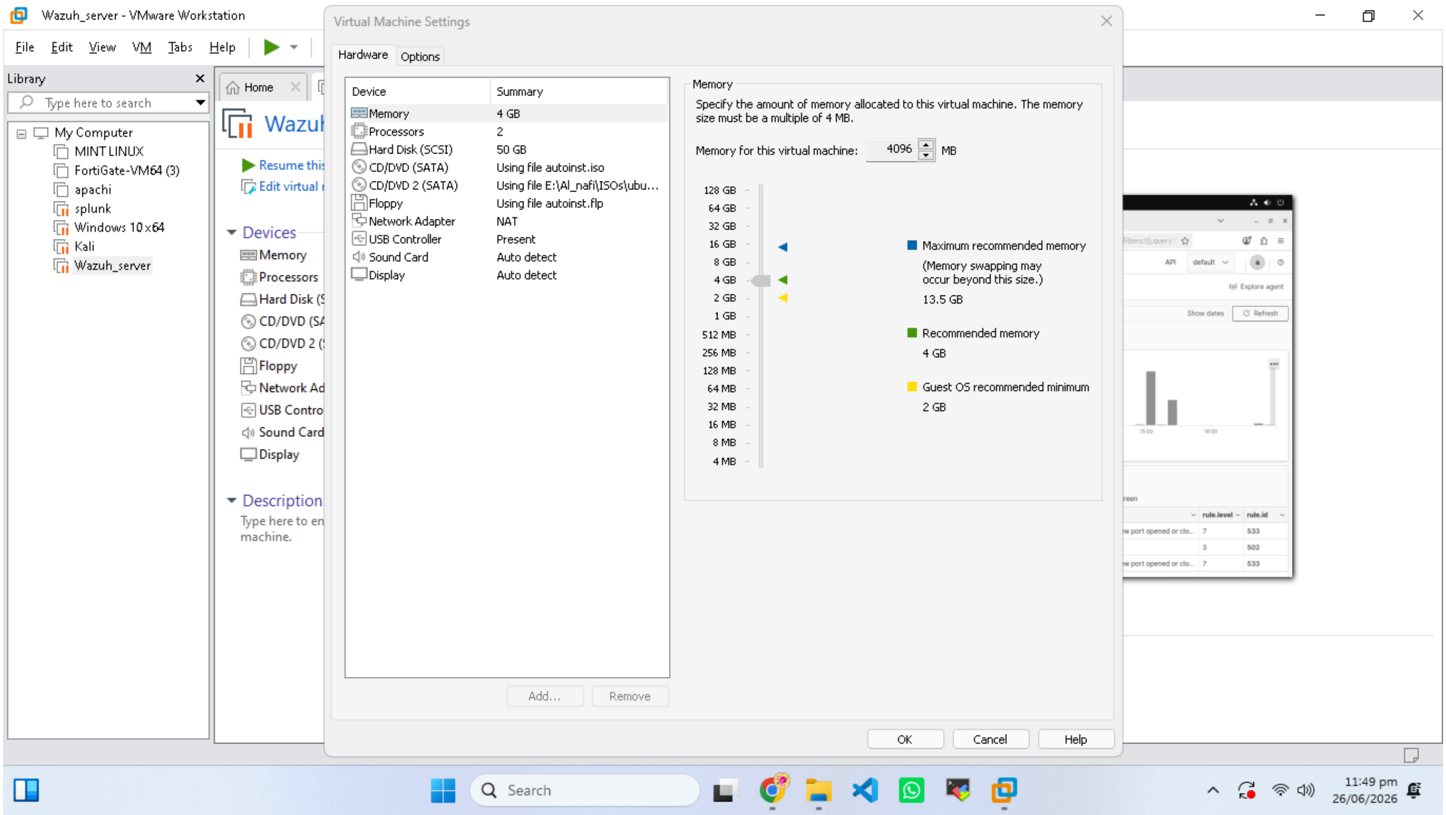
Host System (VMware)
├── Kali Linux (Attacker)
├── Ubuntu Server (Wazuh Manager)
└── Windows 10 (Endpoint)
Network: Host-Only (192.168.147.x) + NAT



Kali VM Specifications - 4GB RAM, 25GB Storage



WINDOWS VM Specifications - 4GB RAM, 50GB Storage



UBUNTU Server VM Specifications - 4GB RAM, 50GB Storage

4.3 VM Specifications

VM	OS	RAM	CPU	Storage	Network	Purpose
Kali Linux	Kali 2024.1	4 GB	2	25 GB	NAT + Host-Only	Attacker
Ubuntu Server	Ubuntu 24.04	8 GB	2	50 GB	NAT + Host-Only	Wazuh Manager
Windows 10	Education	4 GB	2	50 GB	NAT + Host-Only	Endpoint

4.4 IP Configuration

VM	Host-Only IP	NAT IP	Role
Ubuntu Server	192.168.147.137	DHCP	Manager
Kali Linux	192.168.147.129	DHCP	Attacker + Agent
Windows 10	192.168.147.13	DHCP	Agent

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5. WAZUH SIEM INSTALLATION

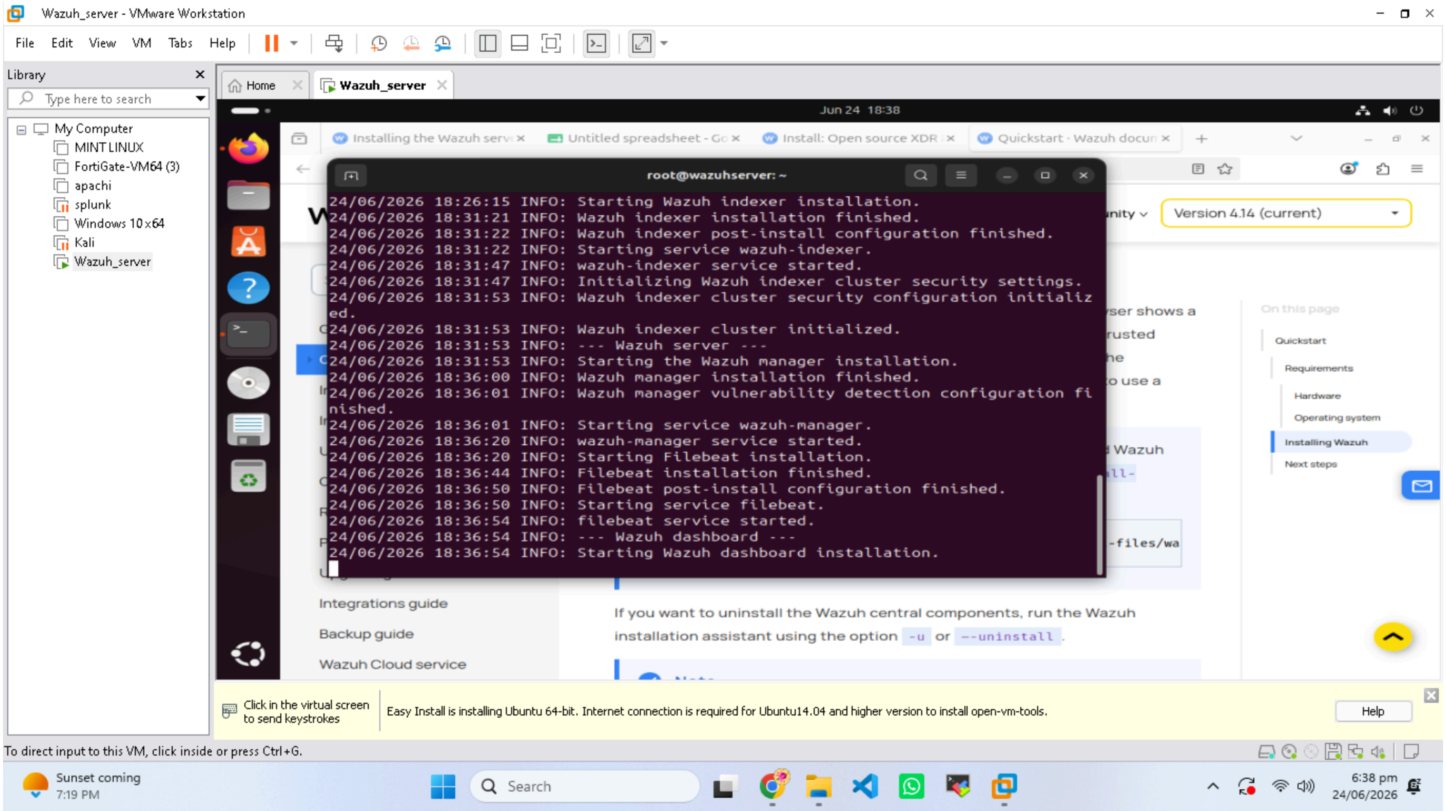
5.1 What is Wazuh?

Wazuh is an open-source SIEM platform for threat detection, integrity monitoring, and compliance.

Agents → Manager → Indexer → Dashboard

5.2 Installation Process

```
curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh
sudo bash ./wazuh-install.sh -a
```

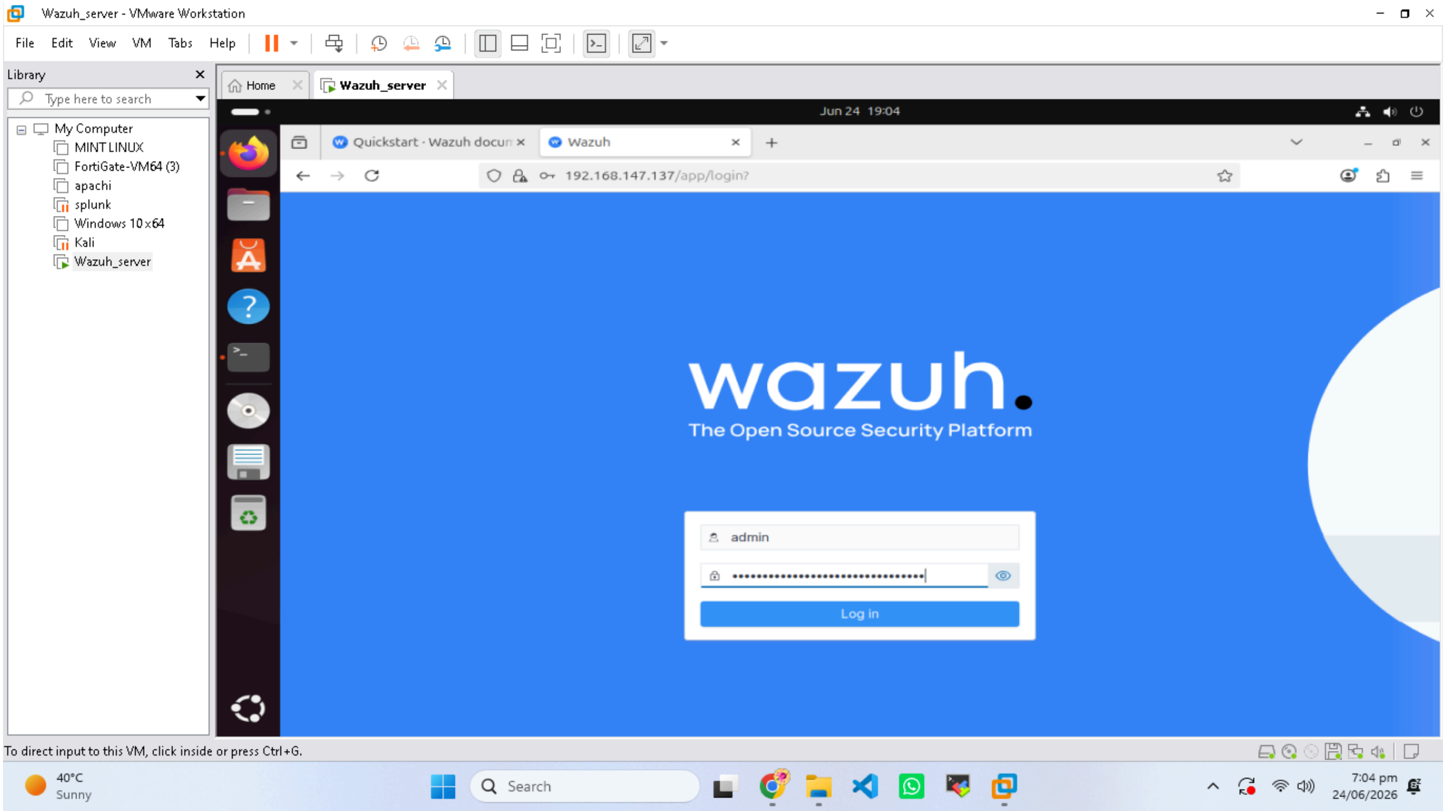


Wazuh Installation Terminal Output

5.3 Post-Installation Verification

Service	Status	Port
wazuh-manager	✅ Active	1514,1515
wazuh-indexer	✅ Active	9200

Login: https://192.168.147.X
Username: admin



Wazuh Dashboard Login Page

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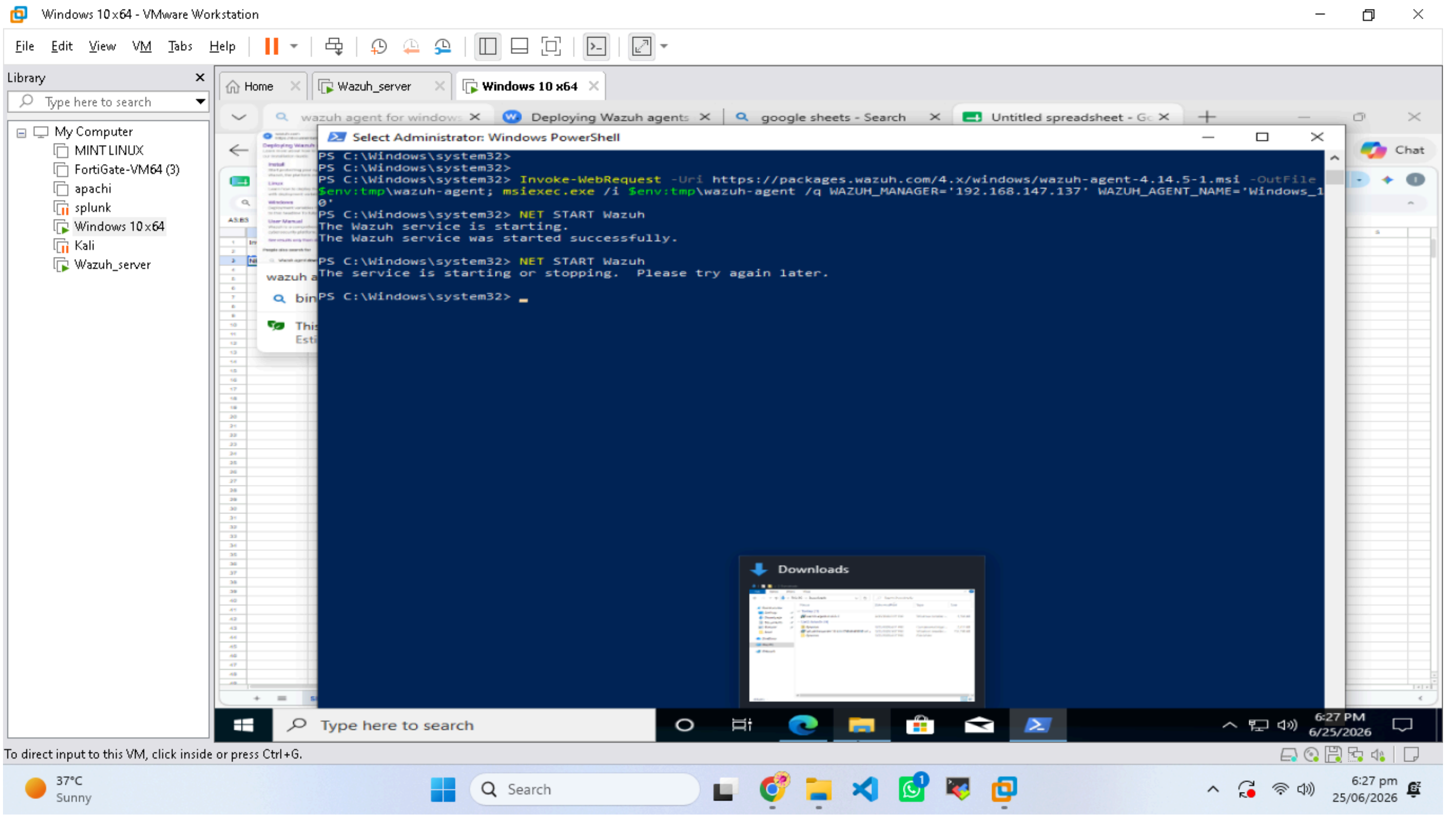
6. WINDOWS AGENT DEPLOYMENT

6.1 Agent Installation

Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.5-1.msi -OutFile wazuh-agent.msi
msiexec.exe /i wazuh-agent.msi /q WAZUH_MANAGER='192.168.147.137' WAZUH_AGENT_NAME='DESKTOP-2GJ375A'
NET START WazuhSvc

6.2 Agent Status

Attribute	Value
Agent Name	DESKTOP-2GJ375A
IP	192.168.147.13
OS	Windows 10
Version	v4.14.5
Status	✅ Active



Windows Agent Active in Dashboard

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7. KALI LINUX AGENT DEPLOYMENT

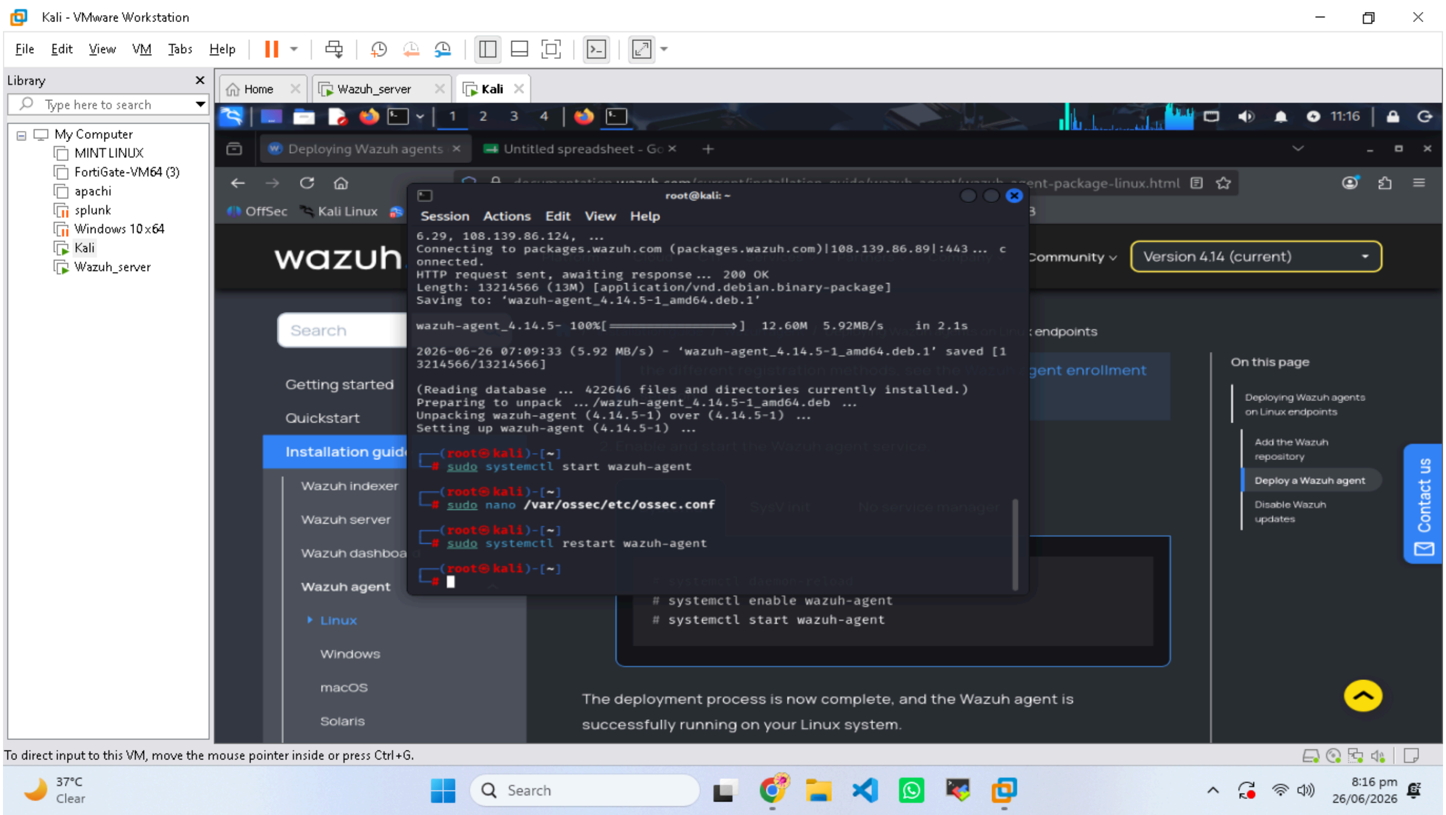
7.1 Agent Installation

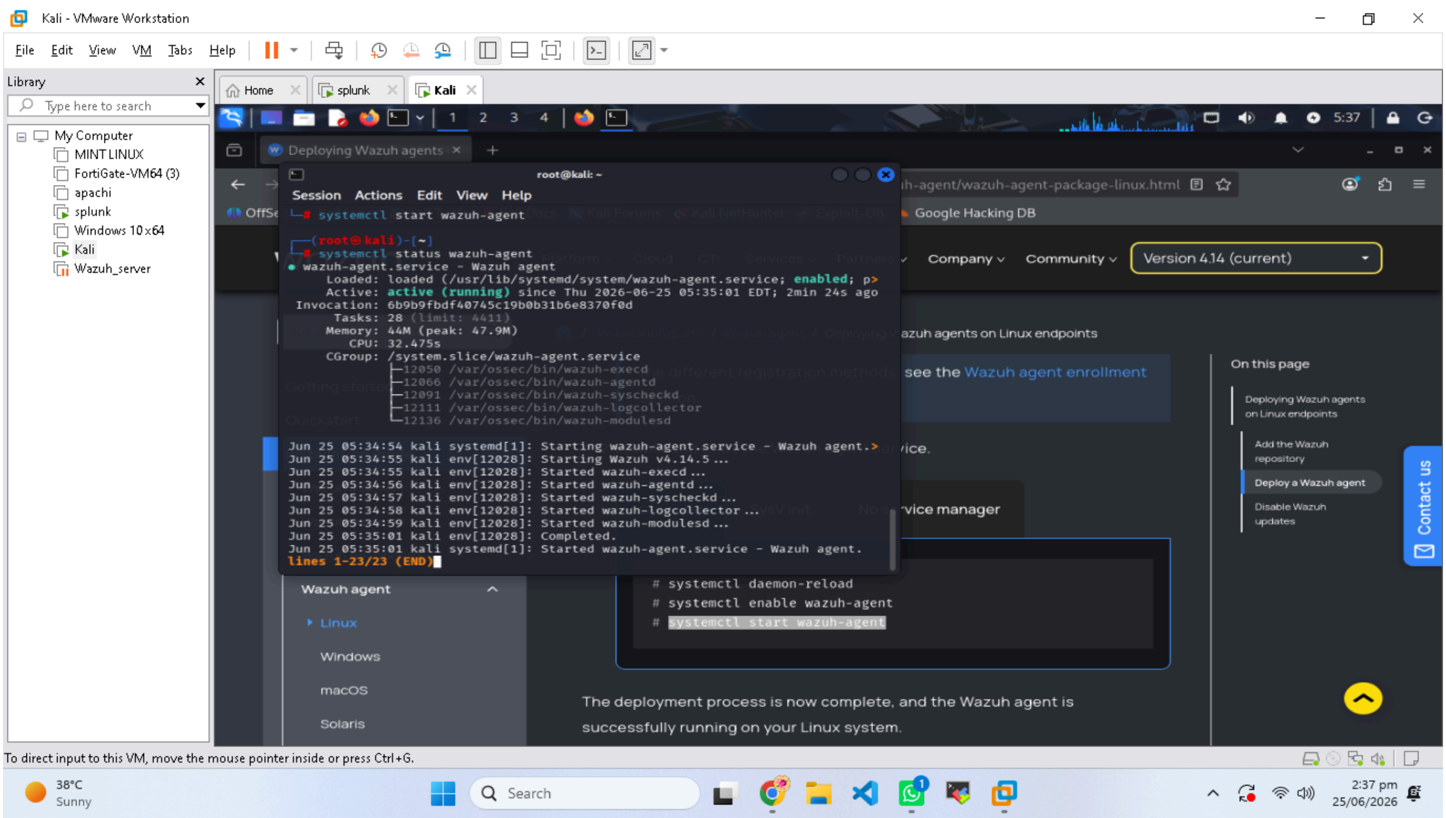
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | gpg --no-default-keyring --keyring gnupg-ring:/usr/share/keyrings/wazuh.gpg --import
echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main" | tee -a /etc/apt/sources.list.d/wazuh.list
apt-get update
WAZUH_MANAGER='192.168.147.137' apt-get install wazuh-agent
systemctl enable wazuh-agent
systemctl start wazuh-agent

7.2 Troubleshooting

Edited /var/ossec/etc/ossec.conf, verified IP, restarted agent, confirmed connectivity.

Attribute	Value
Agent Name	kali
IP	192.168.147.129
Status	✅ Active



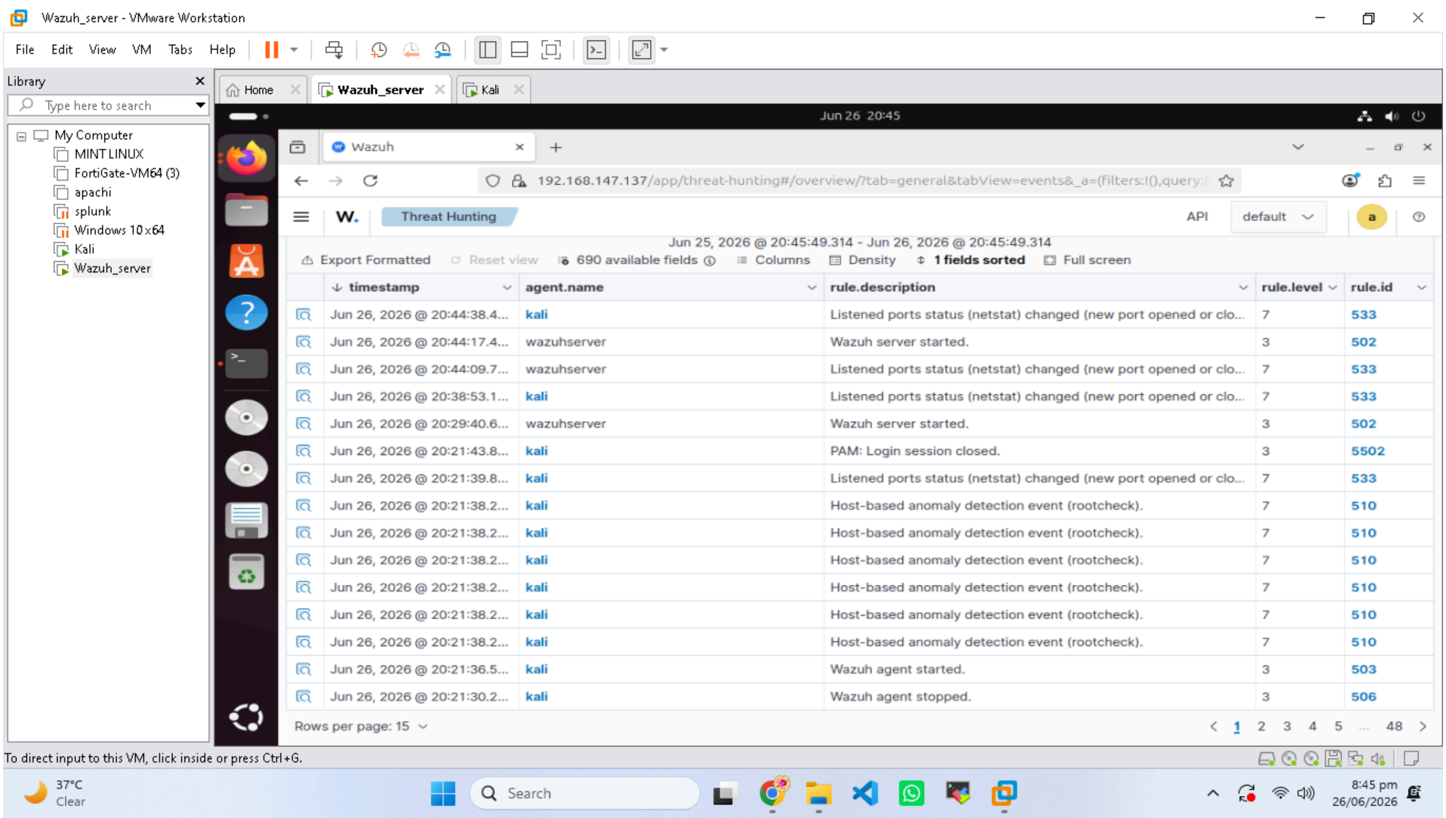


Kali Agent Active in Dashboard

8. ALERT TRIAGE & THREAT HUNTING

8.1 Test Alerts Generated

for i in {1..5}; do ssh wronguser@192.168.147.137; done



Failed SSH Login Alerts

8.2 Alert Triage Table

#	Rule ID	Level	Description	True/False Positive
1	5716	5	SSHD authentication failed	✅ True
2	5712	5	SSH authentication failure	✅ True
3	5710	5	SSH authentication failed	✅ True
4	5740	3	SSH connection closed	❌ False
5	5735	3	SSH connection established	❌ False
6	5502	3	PAM authentication failure	✅ True
7	5711	5	SSH authentication failed	✅ True
8	5730	3	SSH login session closed	❌ False
9	5714	5	SSH authentication failed	✅ True
10	5501	3	PAM authentication error	✅ True

True Positives: 7 **False Positives:** 3

8.3 MITRE ATT&CK Mapping

Technique	MITRE ID	Description
Brute Force	T1110	Adversaries use brute force to gain access
SSH Protocol	T1021.004	Adversaries use SSH for remote access

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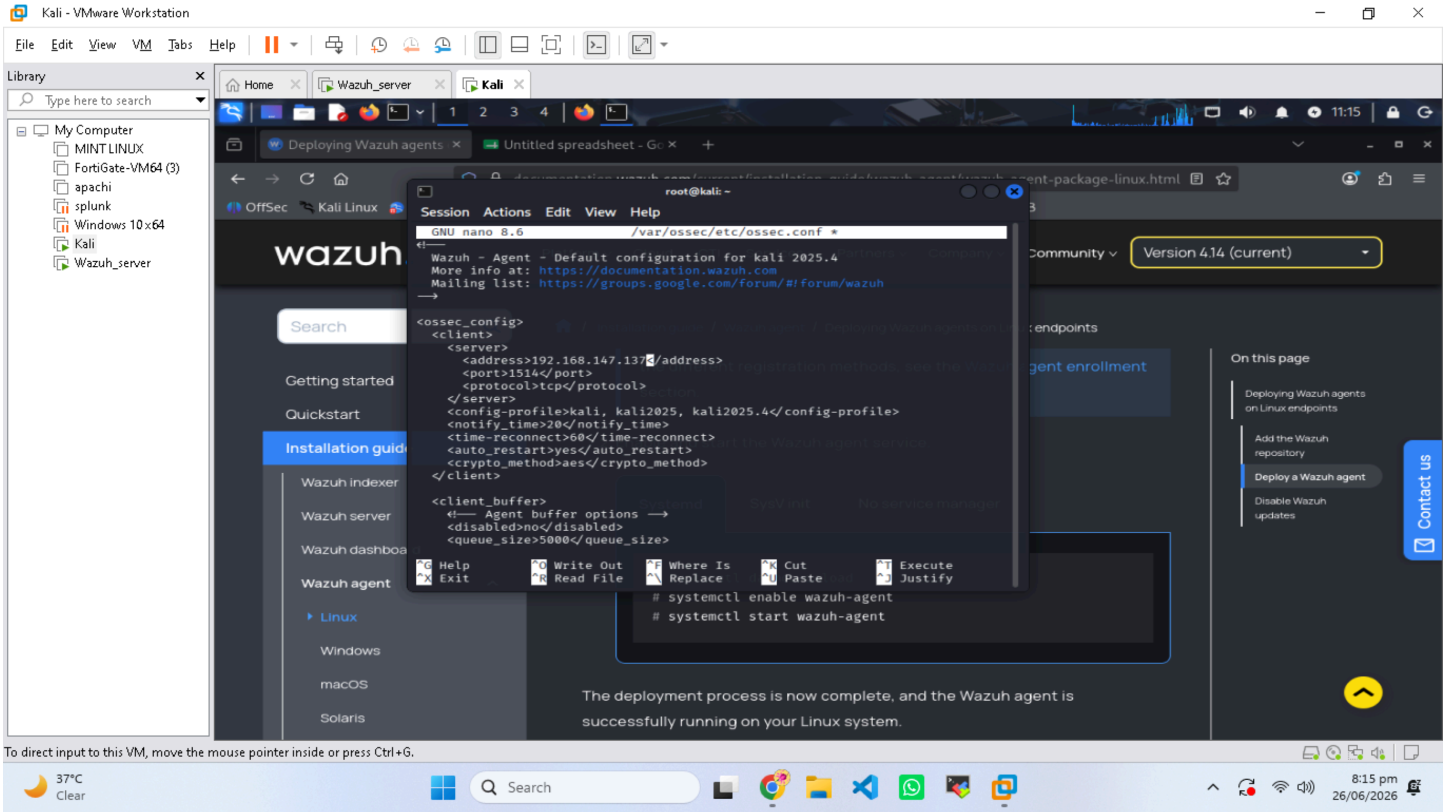
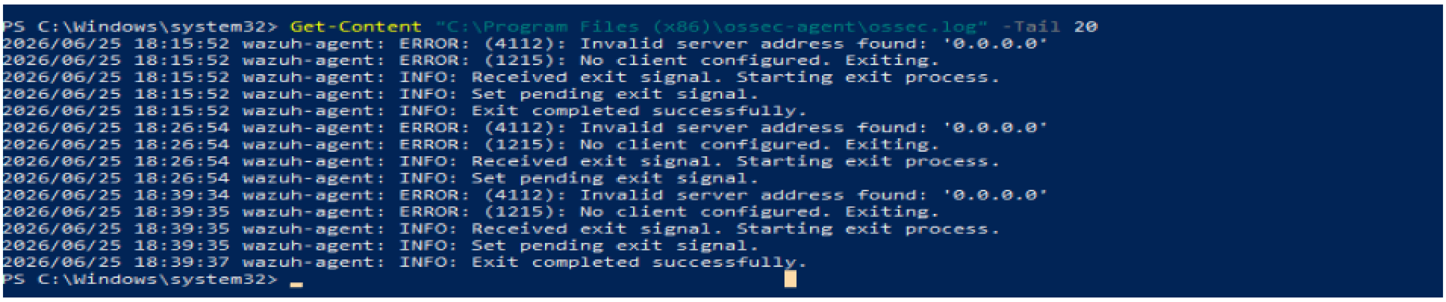
9. CHALLENGES FACED

Challenge 1: Kali Agent “Never Connected” → Fixed by editing config and restarting agent.
Challenge 2: Ubuntu Storage Full → Expanded disk and resized filesystem.
Challenge 3: Windows Firewall Blocking Ping → Enabled ICMP rule.
Challenge 4: Wazuh Dashboard “Not Ready Yet” → Restarted services.
Challenge 5: Permission Issues → Used elevated privileges to edit files.





10. CONCLUSION & KEY LEARNINGS

10.1 What I Learned

#	Topic	Skill Level
1	OSI & TCP/IP Models	Intermediate
2	Ports & Protocols	Practical
3	Wireshark Analysis	Practical
4	Virtualization	Intermediate
5	Wazuh SIEM	Practical
6	Agent Deployment	Practical
7	Alert Triage	Practical
8	MITRE ATT&CK	Beginner

10.2 Skills Gained

Skill	**Before**	**After**
VMware	No Experience	Intermediate
Wazuh SIEM	No Experience	Beginner
Wireshark	No Experience	Practical
Linux CLI	No Experience	Intermediate
Network Troubleshooting	No Experience	Practical
Report Writing	No Experience	Professional

10.3 Key Takeaways

Technical: Built SOC lab, deployed SIEM, analyzed traffic, triaged alerts.
Soft Skills: Problem-solving, documentation, time management, research.
Next Steps: Advanced log correlation, custom rules, simulated attacks, threat intel integration.

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11. REFERENCES & TOOLS

Tools Used

Tool	Purpose
VMware Workstation	Virtualization
Wazuh v4.14.5	SIEM
Wireshark	Traffic Analysis
VirusTotal	Threat Intelligence
Kali Linux	Attack Simulation
Ubuntu Server	SIEM Host
Windows 10	Endpoint
PowerShell	Automation

Final Checklist

Task	Status
Kali VM Setup	✅
Ubuntu VM Setup	✅
Windows 10 VM Setup	✅
Wazuh Installation	✅
Dashboard Access	✅
Windows Agent	✅
Kali Agent	✅
Test Alerts	✅
Alert Triage	✅
Wireshark Analysis	✅